Strassen's Matrix Multiplication.

This algorithm is called strassen's matrix multiplication algorithm and can be accomplished by using the following formula.

$$\begin{bmatrix} C_1 & C_2 \\ C_3 & C_4 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} * \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$$

$$= \begin{bmatrix} m_1+m_4-m_5+m_7 & m_3+m_5 \\ m_2+m_4 & m_1+m_3-m_2+m_6 \end{bmatrix}$$

Where

$m_1 = (A_1 + A_4) * (B_1 + B_4)$
$m_2 = (A_1 + A_4) * B_1$
$m_3 = A_1 * (B_2 - B_4)$
$m_4 = A_4 * (B_3 - B_1)$
$m_5 = (A_1 + A_2) * B_4$
$m_6 = (A_3 - A_1) * (B_1 + B_2)$
$m_7 = (A_2 - A_4) * (B_3 + B_4)$

Analysis: Strarren's algorithm is more efficient than the Conventional Matrix multiplication for sufficiently large values of n, but not for smaller value of n.

The recurence relation for this can be written as:

Consider the relation

$T(n) = 7T(n/2)$

$T(n) = 7(7T(n/2^2))$

$= 7^2 T(n/2^2)$

$= 7^3 T(n/2^3)$

$\vdots$

$= 7^i T(n/2^i)$

$= 7^i T(n/n)$

$= 7^i T(1)$

$= 7^i \cdot 1$

$= 7^{\log_2 n}$

$= n^{\log_2 7}$

$= n^{2.807}$

$T(n) = 7T(n/2)$

Replacing n by $n/2$ in above eqn, we have

$T(n/2) = 7T(n/2^2)$

put $2^i = n$

take both side log

$2^i (\log) = (\log) n$

$\Rightarrow i \log 2 = \log n$

$\Rightarrow i \cdot 1 = \log n$

$i = \log_2 n$

We know

$a^{\log_2 b} = b^{\log_2 a}$

Therefore

Time complexity to multiply two integers

EXAMPLE-1

Let $A = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 7 \\ 1 & 2 \end{bmatrix}$

$m_1 = (A_1 + A_4) * (B_1 + B_4) = (1+6) * (6+4) = 70$
$m_2 = (A_3 + A_4) * B_1 = (5+6) * 6 = 66$
$m_3 = A_1 * (B_2 - B_4) = 1 * (5-4) = 1$
$m_4 = A_1 * (B_3 - B_1) = 6 * (3-6) = -18$
$m_5 = (A_1 + A_2) * B_4 = (1+2) * 4 = 12$
$m_6 = (A_3 - A_1) * (B_1 + B_2) = (5-1) * (6+5) = 44$
$m_7 = (A_2 - A_1) * (B_3 + B_4) = (2-6) * (3+4) = -28$

$$C = \begin{bmatrix} 70-18-12-28 & 1+12 \\ 66-18 & 70+1-66+44 \end{bmatrix}$$

$$C = \begin{bmatrix} 12 & 13 \\ 48 & 49 \end{bmatrix}$$

EXAMPLE-2

Using divide and Conquer method, multiply the following two matrices with the help of Strassen's matrix multiplication.

$$\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 8 & 7 \\ 1 & 2 \end{bmatrix}$$

Solution: let $A = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 8 & 7 \\ 1 & 2 \end{bmatrix}$.

So $A_1 = 1$ | $B_1 = 8$
$A_2 = 2$ | $B_2 = 7$
$A_3 = 5$ | $B_3 = 1$